BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

BOTANY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Multiple Choice Questions)

Note: Q.1 Choose the correct answer for each from the given options:

Q1. A _____ which updates itself perfectly to conditions of environment is termed as:

(a) Succession
(b) Xerosere
(c) Clusius
(d) None of these

Q2. The nutrients which are necessary for the operatio of ecosystem are:

(a) Micro - Nutrients
(b) Macro – Nutrients
(c) Trace - Element
(d) Operation – Nutrients

Q3. A phenomenon of inheritance in which alleles of contrasting characters are not completely domination over other:

(a) Complete Dominance
(b) Incomplete Dominance
(c) Intermediate – Dominance
(d) None of these

Q4. Cells in the spaces between the fingers that are no longer needed perform.

(a) Apoptosis
(b) Growth
(c) Mitosis
(d) Necrosis

Q5. Vertically sea is divided in to _____ Zones.

(a) Euphotic, bathyal and abysal
(b) Bathyal and abyssal
(c) Euphotic and aphotic
(d) Neritic and oceanic

Q6. The nature of endosperm (food storage tissue is _____

(a) N
(b) 2 N
(c) 3 N
(d) 4 N

Q7. Which one is the true statement for "Translocation" regarding chromosomal aberration?

(a) A segment of chromosome is invented
(b) A segment of chromosome missing
(c) A segment of chromosome present in excess
(d) Transfer of segment of chromosome to non – homologous chromosome.

Q8. Plasmolysis can occur, when a plant cell is placed in _____ solution.

(a) Isotonic
(b) Hypertonic
(c) Hypotonic
(d) None of these

Q9. The relationship between members of two different species in which both get benefit.

(a) Symbiosis
(b) Mutualism
(c) Parasitism
(d) Commensalism

Q10. In plant cell, the correct water potential of solution in vacuole is:

(a) Water – Potential
(b) Osomis
(c) Solute – Potential
(d) Osmotic - Pressure

Q11. The movement which occur due to difference in the rate of growth on the two opposite sides of a plant organ is called:

(a) Tropic
(b) Tactic
(c) Nostic
(d) All of these

Q12. A process in which a part of plant develops into a new plant:

(a) Clonning
(b) Apomixis
(c) Tissue culture
(d) Vegetative Propagation

Q13. Genetic –code is command from nucleus to ectoplasm in the form of:

(a) Anti – Codon
(b) Codon
(c) m – RNA
(d) t – RNA

Q14. Parthenocarpy is the type of asexual reproduction.

(a) Type of asexual reproduction
(b) Fruit form without fertilization
(c) Found in plants
(d) All of these

Q15. The tubular lignified dead – cells with large hollow cavities and oblique transverse wall called:

(a) Fibres
(b) Tracheae
(c) Trachieds
(d) Scieroids

Q16. The curvature movement of plant in response to touch stimulus.

(a) Haptonastic
(b) Epinastic
(c) Sesmonastic
(d) Thigmo – Tropism

Q17. One complete turn of DNA contains:

(a) 20 Nucleotides
(b) 10 Nucleotides
(c) 5 Nucleotides
(d) 2 Nucleotides

SECTION – B (SHORT ANSWER)

> Note: Answer any EIGHT of the following questions. Each question carries 02 marks.

Q2. Which of the phyto- harmone is considered as growth inhibitor and why?

Q3. How Herbivory is stress for plants and how plants cope from this stress?

Q4. Why "Hetero – Chromatin" part of chromosome is never expressed?

Q5. Why Ethene is considered as fruit ripening hormone and how does ethane ripens the fruit?

Q6. Why coniferous forests are named so and where these forests are located in Pakistan?

Q7. Why secondary succession is more repid than primary succession?

Q8. On the basis of position of centromere enlist the four types of chromosomes.

Q9. Distinguish between any ONE of the following in tabulated form: (i) Klinefether's Syndrome and Turner's Syndrome (ii) Vascular Cambium and Cork Cambium

Q10. Why Rh incompatibility could be a danger to the developing foetus and mother?

Q11. Define any TWO of the following. (i) Topography (ii) Tundra (iii) Mitotic Apparatus (iv) Corymb

Q12. What is ecological Niche?

Q13. Why abnormal daughter cells are produced during cell division?

SECTION – C (DESCRIPTIVE ANSWER)

> Note: Answer any TWO of the following questions. Each question carries 05 marks.

Q14. What is mono hybrid cross? Describe the law of inheritance which explain the inheritance of mono hybrid cross?

Q15. Describe the experiment of Griffith, Hershy and Chase to prove that "DNA is the hereditary material".

Q16. What do you mean by karyokinesis? Describe various stages of karyokinesis of somatic cell division.